

Lab 3 Architecture Justification

Name: Shashank D

SRN: PES1UG24CS433

System: Academic Elective Bidding and Allocation System

Architecture Selection

We chose Layered Architecture for the Academic Elective Bidding and Allocation System. The design has a Presentation Layer, Business Logic Layer and Data Layer. The component diagram separates the portal, controller, authentication, course configuration, bidding, allocation, results and storage responsibilities.

Architectural Style Analysis

Layered architecture is suitable because it separates academic rules from the web interface and database. Its main drawback is coupling between layers. Microservices allow independent deployment and scaling but add network calls and distributed data consistency problems. Client-server provides centralized control but may create a server bottleneck; it does not specify how the server is divided internally.

Two Reasons

1. The bidding service can enforce the 100-credit total, prerequisites and deadline independently of the UI. The allocation engine handles seat limits, CGPA tie-breakers and timetable clashes. This separation makes each set of academic rules easier to maintain.
2. Bid updates and allocation results need consistent data. A shared academic database supports atomic bid changes and a fixed snapshot after the deadline, so the solver works with a complete, stable set of bids.

Security Advantage

The authentication service verifies login and roles. The controller and business services restrict course configuration and manual overrides to the Academic Registrar and limit students to their own records. The UI has no direct database access. Bid modifications and overrides are saved with audit entries containing the user, timestamp and prior state; restricted database permissions prevent audit updates or deletion.

Performance Benefit

Business components communicate through local API calls. The allocation engine reads a deadline-locked snapshot and runs as a background job, keeping the portal responsive. Indexed course, bid and slot data reduce repeated lookups. Processing 5,000 bids across 40 electives in under 30 seconds is the specified target and must be verified by benchmarking.